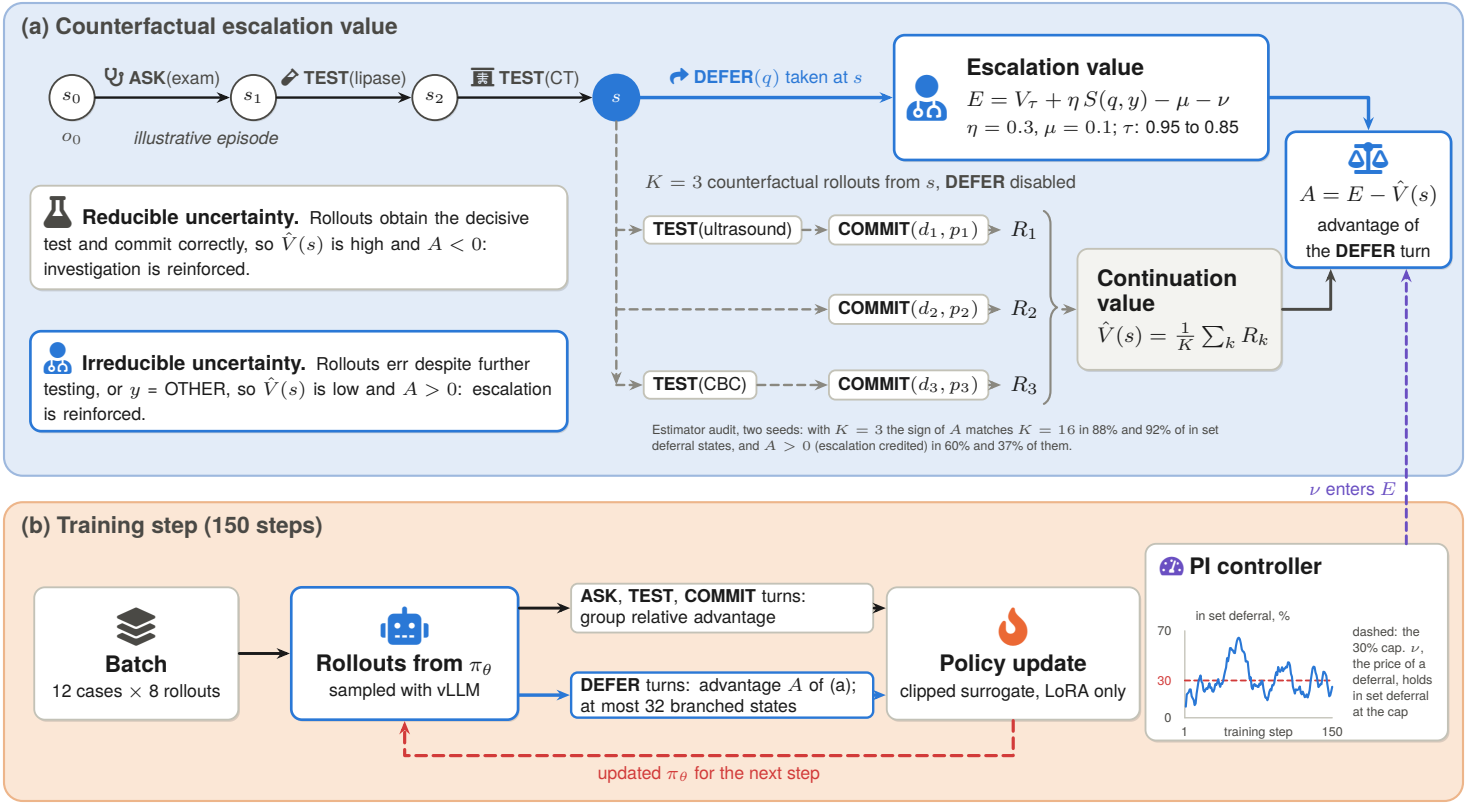
(a) Counterfactual escalation value
s_0
ASK(exam)
s_1
TEST(lipase)
s_2
TEST(CT)
s
o_0
illustrative episode
DEFER(q) taken at s
Escalation value
$E = V_\tau + \eta\, S(q,y) - \mu - \nu$
$\eta = 0.3$, $\mu = 0.1$; τ: 0.95 to 0.85
$A = E - \hat{V}(s)$
advantage of the DEFER turn
Reducible uncertainty. Rollouts obtain the decisive test and commit correctly, so $\hat{V}(s)$ is high and $A < 0$: investigation is reinforced.
Irreducible uncertainty. Rollouts err despite further testing, or y = OTHER, so $\hat{V}(s)$ is low and $A > 0$: escalation is reinforced.
$K = 3$ counterfactual rollouts from s, DEFER disabled
TEST(ultrasound)
COMMIT(d_1, p_1)
R_1
COMMIT(d_2, p_2)
R_2
TEST(CBC)
COMMIT(d_3, p_3)
R_3
Continuation value
$\hat{V}(s) = \frac{1}{K}\sum_k R_k$
Estimator audit, two seeds: with $K = 3$ the sign of A matches $K = 16$ in 88% and 92% of in set deferral states, and $A > 0$ (escalation credited) in 60% and 37% of them.
ν enters E
(b) Training step (150 steps)
Batch
12 cases × 8 rollouts
Rollouts from π_θ
sampled with vLLM
ASK, TEST, COMMIT turns: group relative advantage
DEFER turns: advantage A of (a); at most 32 branched states
Policy update
clipped surrogate, LoRA only
PI controller
in set deferral, %
70
30
0
1
training step
150
dashed: the 30% cap. ν, the price of a deferral, holds in set deferral at the cap
updated π_θ for the next step